

EE708: Fundamentals of Data Science and Machine Intelligence

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Module 3B Regression Analysis and Modeling: Bayesian Linear Regression, Bias-Variance Decomposition

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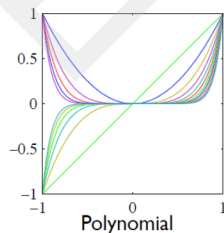
Linear Basis Function Models

- Linear regression extended to consider fixed basis functions:

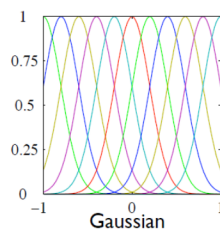
$$y(\mathbf{w}) = \sum_{j=0}^{M-1} w_j \phi_j(x) = \mathbf{w}^T \boldsymbol{\phi}(x)$$

Where $\mathbf{w} = [w_0, w_1, \dots, w_{M-1}]^T$ are weights and $\boldsymbol{\phi} = [\phi_0, \phi_1, \dots, \phi_{M-1}]^T$ are basis functions.

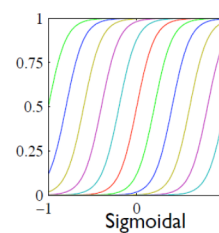
- Possible basis functions include polynomials, Fourier, wavelet, and many others.



$$\phi_j(x) = x^j$$



$$\phi_j(x) = \exp\left\{-\frac{(x - \mu_j)^2}{2s^2}\right\}$$



$$\phi_j(x) = \sigma\left(\frac{x - \mu_j}{s}\right)$$

where $\sigma(a) = \frac{1}{1 + \exp(-a)}$

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Maximum Likelihood and Least Squares

- Presume target t is generated by a deterministic function $y(x, \mathbf{w})$ with additive Gaussian noise ϵ so that

$$t = y(x, \mathbf{w}) + \epsilon$$

where ϵ is zero mean Gaussian random variable with precision (inverse variance) β , $\epsilon \sim \mathcal{N}(0, \beta^{-1})$

$$t \sim \mathcal{N}(y(x, \mathbf{w}), \beta^{-1})$$

$$p(t|x, \mathbf{w}, \beta) = \mathcal{N}(t|y(x, \mathbf{w}), \beta^{-1})$$

- With Gaussian conditional distribution conditional mean is:

$$\mathbb{E}(t|x) = \int t p(t|x) dt = y(x, \mathbf{w})$$

- With a set of input points $\mathbf{X} = \{x_n\}_{n=1}^N$ independently drawn:

$$p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \boldsymbol{\phi}(x_n), \beta^{-1})$$

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Maximum Likelihood and Least Squares

- Log likelihood:**

$$\ln p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \sum_{n=1}^N \ln \mathcal{N}(t_n | \mathbf{w}^T \boldsymbol{\phi}(x_n), \beta^{-1})$$

$$\text{Using } \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

$$\ln \mathcal{N}(t_n | \mathbf{w}^T \boldsymbol{\phi}(x_n), \beta^{-1}) = \frac{1}{2} \ln \beta - \frac{1}{2} \ln 2\pi - \frac{\beta}{2} (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2$$

$$\sum_{n=1}^N \ln \mathcal{N}(t_n | \mathbf{w}^T \boldsymbol{\phi}(x_n), \beta^{-1}) = \sum_{n=1}^N \left(\frac{1}{2} \ln \beta - \frac{1}{2} \ln 2\pi - \frac{\beta}{2} (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2 \right)$$

$$\ln p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) - \frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2$$

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Maximum Likelihood and Least Squares

- Gradient:

$$\ln p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) - \frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2$$

$$\nabla \ln p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n)) \boldsymbol{\phi}(x_n)^T$$

- Set gradient to 0

$$\sum_{n=1}^N t_n \boldsymbol{\phi}(x_n)^T - \mathbf{w}^T \sum_{n=1}^N \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^T = 0$$

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Maximum Likelihood and Least Squares

- Solving for weights yields normal equations for least squares:

$$\mathbf{w}_{ML} = (\boldsymbol{\Phi}^T \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \mathbf{t}$$

- $\boldsymbol{\Phi} = \begin{bmatrix} \phi_0(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \cdots & \phi_{M-1}(\mathbf{x}_2) \\ \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{bmatrix}$ is the design matrix

- $(\boldsymbol{\Phi}^T \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \mathbf{t}$ is the pseudo-inverse of $\boldsymbol{\Phi}$

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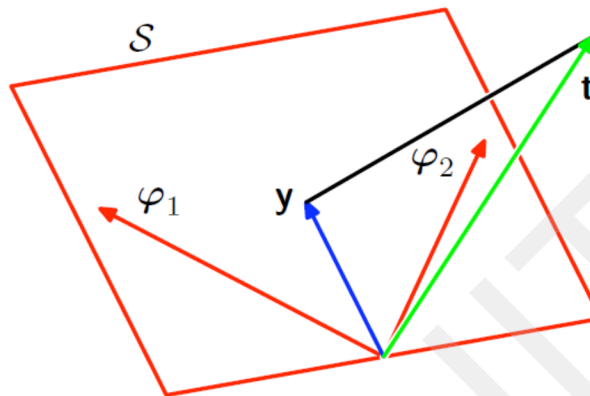
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Geometry of Least Squares

- Least-squares regression is obtained by finding the orthogonal projection of the data vector $\mathbf{t}$ onto the subspace spanned by the basis functions.
- Intuition: Sum of squares error is 1/2 squared Euclidean distance between $\mathbf{y}$ and $\mathbf{t}$. Thus least-squares solution would move $\mathbf{t}$ as close as possible to $\mathbf{y}$ in the subspace S .



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Online Learning

- For large datasets may need to learn sequentially on sequences of smaller datasets, summing error

$$E = \sum_n E_n$$

- Sequential gradient descent (also called stochastic gradient descent):

$$\mathbf{w}^{(\tau+1)} = \mathbf{w}^{(\tau)} - \eta \nabla E_n$$

where τ is the iteration number and η is the learning rate.

- For sum-of-squares we get Least Mean Squares (LMS):

$$\mathbf{w}^{(\tau+1)} = \mathbf{w}^{(\tau)} + \eta \left(t_n - \mathbf{w}^{(\tau)T} \phi_n \right) \phi_n$$

- Learning rate must be chosen carefully.

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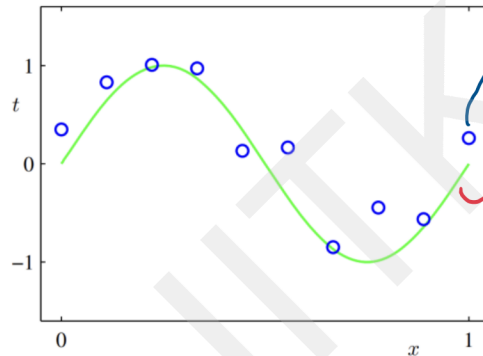
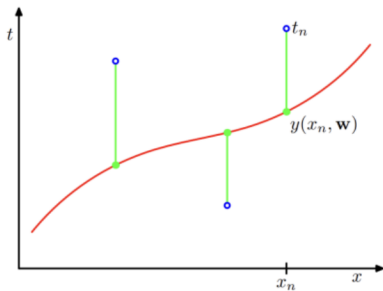
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Overfitting

$$y(x, \mathbf{w}) = w + 0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

$$g(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y(x, \mathbf{w}) - t_n)^2$$

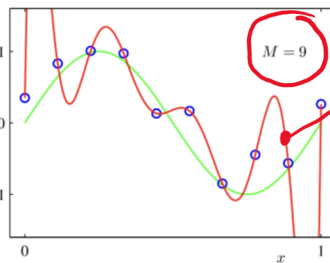
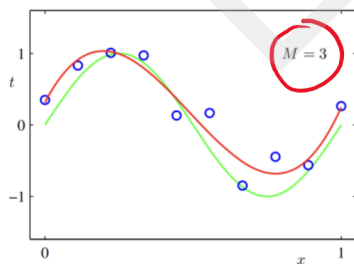
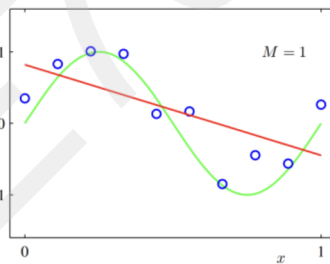
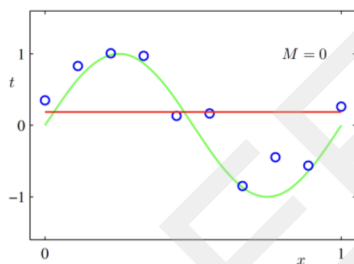


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Overfitting: Varying M



	M = 0	M = 1	M = 6	M = 9
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

Large values!

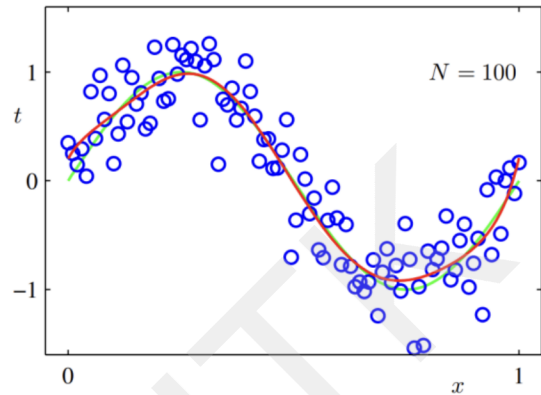
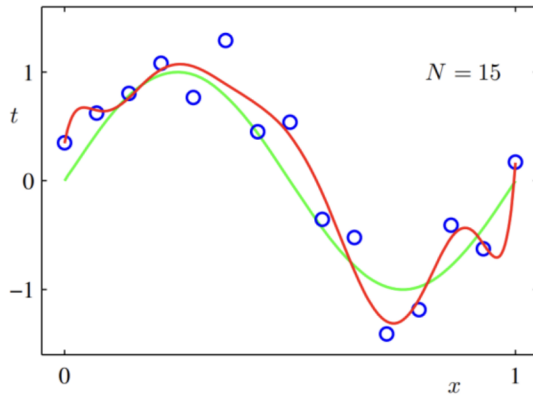
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Overfitting: Effect of Increasing Data Size



$M=9$
varying N

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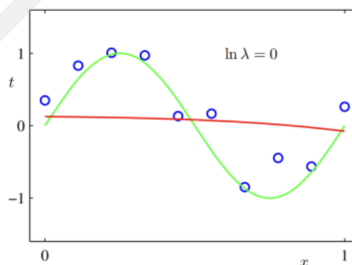
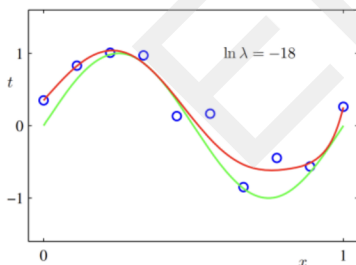
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Regularization

$$\tilde{g}(w) = \frac{1}{2} \sum_{n=1}^N (y(x, w) - t_n)^2 + \underbrace{\frac{\lambda}{2} \|w\|^2}_{\text{Regression penalty term}}$$

$M=9$



⊛ Note 'shrinkage' of w 's (reduction)

	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
w_0^*	0.35	0.35	0.13
w_1^*	232.37	4.74	-0.05
w_2^*	-5321.83	-0.77	-0.06
w_3^*	48568.31	-31.97	-0.05
w_4^*	-231639.30	-3.89	-0.03
w_5^*	640042.26	55.28	-0.02
w_6^*	-1061800.52	41.32	-0.01
w_7^*	1042400.18	-45.95	-0.00
w_8^*	-557682.99	-91.53	0.00
w_9^*	125201.43	72.68	0.01

unregularized
 $\ln \lambda = -\infty$ Good ✓

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Regularized Least Squares

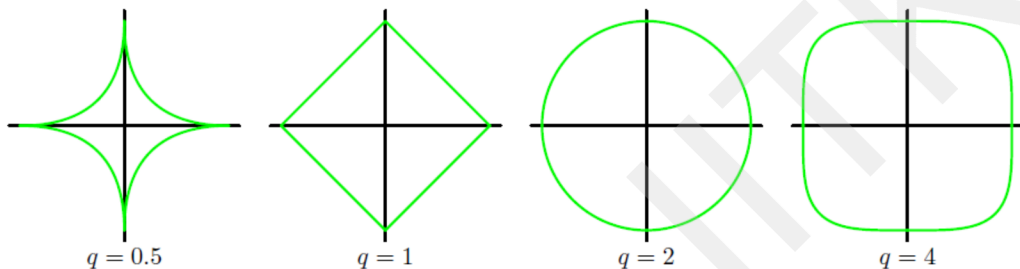
- Updated cost function with general regularization with q is given by

$$\frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2 + \frac{\lambda}{2} \sum_{j=1}^M |\mathbf{w}_j|^q$$

- For $q = 2$ (Ridge Regression)

- updated cost function is $\frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$
- Setting gradient to 0 yields extension of least squares solution

$$\mathbf{w} = (\lambda \mathbf{I} + \boldsymbol{\Phi}^T \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \mathbf{t}$$



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Regularized Least Squares

- Regularized cost function

$$\frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2 + \frac{\lambda}{2} \sum_{j=1}^M |\mathbf{w}_j|^q$$

- Minimizing regularized cost function is equivalent to:

$$\min \frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2$$

Subject to, $\sum_{j=1}^M |\mathbf{w}_j|^q \leq \eta$

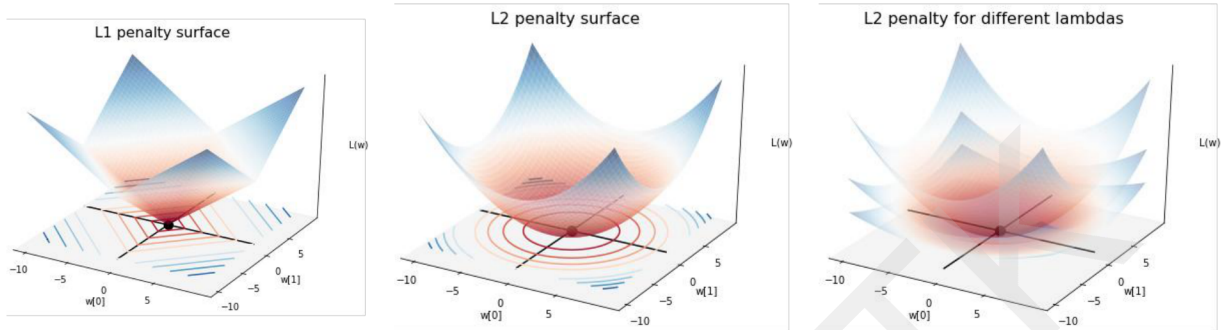
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Regularized Least Squares



Loss Surfaces with Regularization (Lowest Lambda is the Flattest)

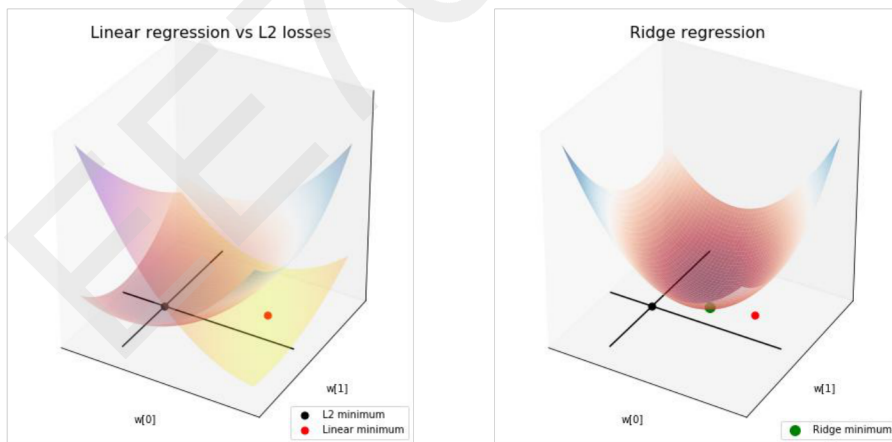
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Regularized Least Squares



Linear and L2 Loss Plots, Plot of Ridge = Linear + L2 : Comparison of Surface Plots

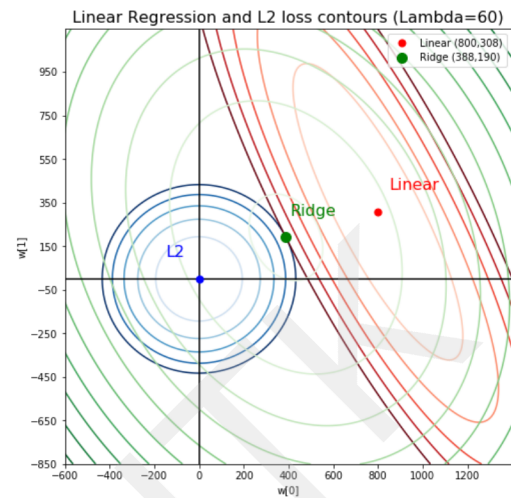
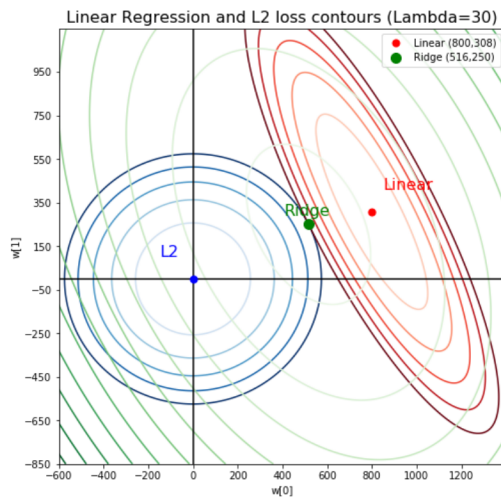
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Regularized Least Squares



Linear and L2 Loss Plots, Plot of Ridge = Linear + L2: Comparison of Contour Plots for various Lambda

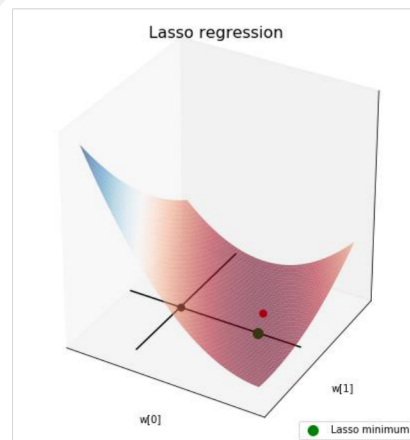
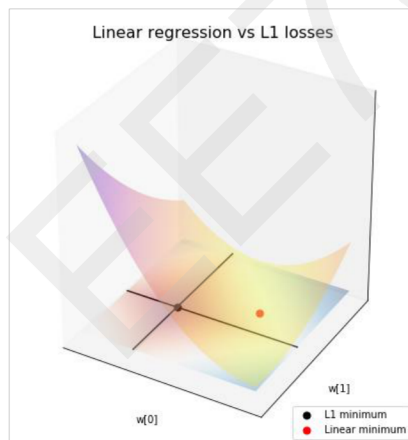
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Regularized Least Squares



Linear and L1 Loss Plots, Plot of Ridge = Linear + L1 : Comparison of Surface Plots

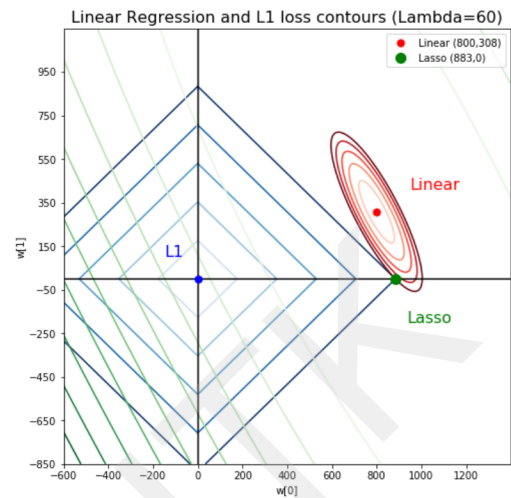
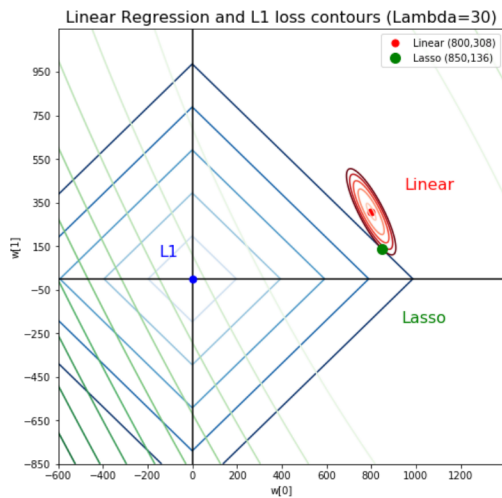
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Regularized Least Squares



Linear and L1 Loss Plots, Plot of Ridge = Linear + L2: Comparison of Contour Plots for various Lambda

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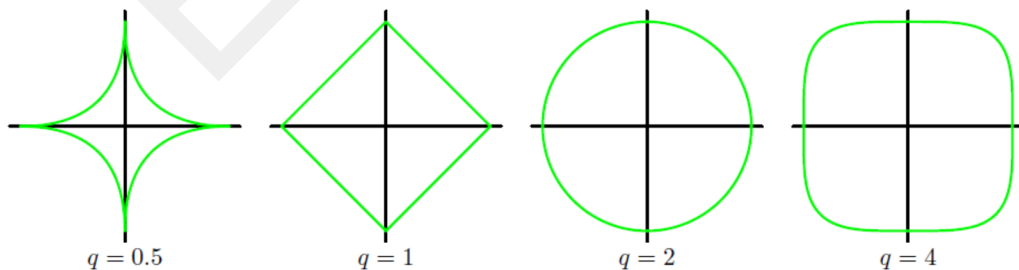
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Regularized Least Squares

- Cost function with general regularization with q is given by

$$\frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \boldsymbol{\phi}(x_n))^2 + \frac{\lambda}{2} \sum_{j=1}^M |\mathbf{w}_j|^q$$

Loss Surfaces with Regularization



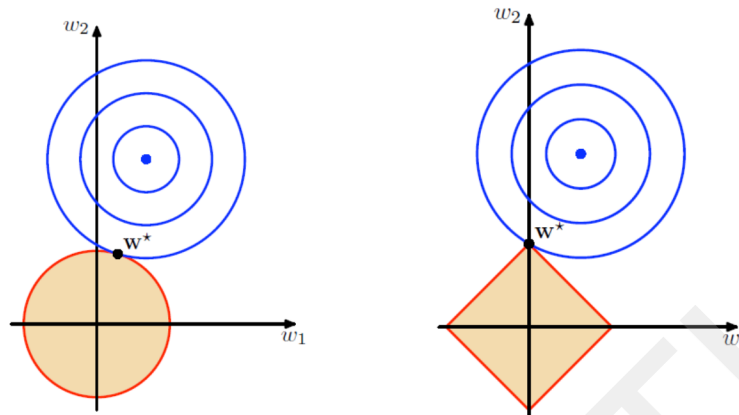
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Visualization of Regularized Least Squares



Plot of contours of unregularized error function along with constraint region on the quadratic regularizer (left, $q=2$) versus lasso regularizer (right, $q=1$). For lasso, a sparse solution is generated with $w_1 = 0$

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Bias-Variance Decomposition (Bayesian view of Fixing λ)

• How to best set the λ parameter for regularization?

• Conditional expectation:

$$h(x) = \mathbb{E}[t|x] = \int t p(t|x) dt$$

• Expected squared loss written with noise as second term:

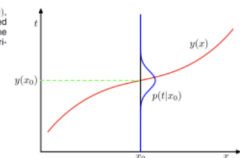
$$\mathbb{E}[L] = \int (y(x) - h(x))^2 p(x) dx + \int (h(x) - t)^2 p(x, t) dx dt$$

• We will minimize the first term. **But we cannot hope to ever know the perfect regression function $h(x)$**

• In a Bayesian model uncertainty is expressed as posterior over $\mathbf{w}$

• In frequentist treatment we make a point-estimate of $\mathbf{w}$. Assess confidence by making predictions over subsets of data and taking mean performance.

The regression function $y(x)$, which minimizes the expected squared loss, is given by the mean of the conditional distribution $p(t|x)$.



derivation
next pps

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Expected Squared Loss

- Average or expected loss

$$\mathbb{E}[L] = \int \int L(t, y(\mathbf{x})) p(\mathbf{x}, t) d\mathbf{x} dt$$

- Common choice of loss function in regression problem in square loss:

$$\mathbb{E}[L] = \int \int (y(\mathbf{x}) - t)^2 p(\mathbf{x}, t) d\mathbf{x} dt$$

- Goal is to choose $y(\mathbf{x})$ that minimizes $\mathbb{E}[L]$.

$$\frac{d\mathbb{E}[L]}{dy(\mathbf{x})} = 2 \int (y(\mathbf{x}) - t) p(\mathbf{x}, t) dt = 0$$

- Solving for $y(\mathbf{x})$, and using sum and product rules of probability,

$$y(\mathbf{x}) = \frac{\int t p(\mathbf{x}, t) dt}{p(\mathbf{x})} = \int t p(t|\mathbf{x}) dt = \mathbb{E}_t[t|\mathbf{x}]$$

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Expected Squared Loss

Another approach to minimize $\mathbb{E}[L]$:

$$\begin{aligned} (y(\mathbf{x}) - t)^2 &= (y(\mathbf{x}) - \mathbb{E}_t[t|\mathbf{x}] + \mathbb{E}_t[t|\mathbf{x}] - t)^2 \\ &= (y(\mathbf{x}) - \mathbb{E}_t[t|\mathbf{x}])^2 + 2(y(\mathbf{x}) - \mathbb{E}_t[t|\mathbf{x}])(\mathbb{E}_t[t|\mathbf{x}] - t) + (\mathbb{E}_t[t|\mathbf{x}] - t)^2 \end{aligned}$$

Substituting into loss function and performing integration over t :

$$\mathbb{E}[L] = \int (y(\mathbf{x}) - \mathbb{E}_t[t|\mathbf{x}])^2 p(\mathbf{x}) d\mathbf{x} + \int (\mathbb{E}_t[t|\mathbf{x}] - t)^2 p(\mathbf{x}) d\mathbf{x}$$

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Bias-Variance Decomposition

- Take integrand of first term using some subset of data $\mathcal{D}$. $(y(x, \mathcal{D}) - h(x))^2$ which varies with data, thus take its mean
- Add and subtract expected value for the data

$$\begin{aligned} & (y(x, \mathcal{D}) - \mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})) + \mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})) - h(x))^2 \\ &= (y(x, \mathcal{D}) - \mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})))^2 + (\mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})) - h(x))^2 \\ &+ 2(y(x, \mathcal{D}) - \mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D}))) (\mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})) - h(x)) \end{aligned}$$

- Take expectation with respect to $\mathcal{D}$; final term vanishes

$$\mathbb{E}_{\mathcal{D}} \left[(y(x, \mathcal{D}) - h(x))^2 \right] = \left(\mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})) - h(x) \right)^2 + \mathbb{E}_{\mathcal{D}} \left((y(x, \mathcal{D}) - \mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})))^2 \right)$$

- **First term is bias: extent to which average prediction differs from desired regression function**
- **Second term is variance: extent to which individual solutions vary around the average. Thus measures sensitivity to data.**

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Bias-Variance Decomposition

- expected loss = $(bias)^2 + \text{variance} + \text{noise}$,
where:

$$(bias)^2 = \int \left(\mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})) - h(x) \right)^2 p(x) dx$$

$$variance = \int \mathbb{E}_{\mathcal{D}} \left((y(x, \mathcal{D}) - \mathbb{E}_{\mathcal{D}}(y(x, \mathcal{D})))^2 \right) p(x) dx$$

$$noise = \int (h(x) - t)^2 p(x) dx$$

- Very flexible models have low bias and high variance
- Relatively rigid models have high bias and low variance
- Optimal model balances the two

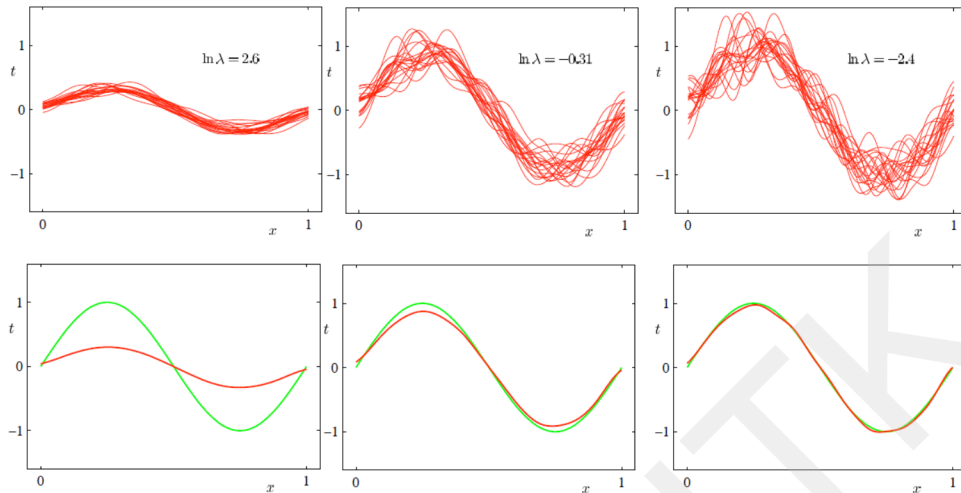
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Example



100 datasets each with 25 data points. Fit with 25 Gaussian basis functions. Regularization parameter λ is varied. Top are individual fits. Bottom is average fit along with generating sine function in green.

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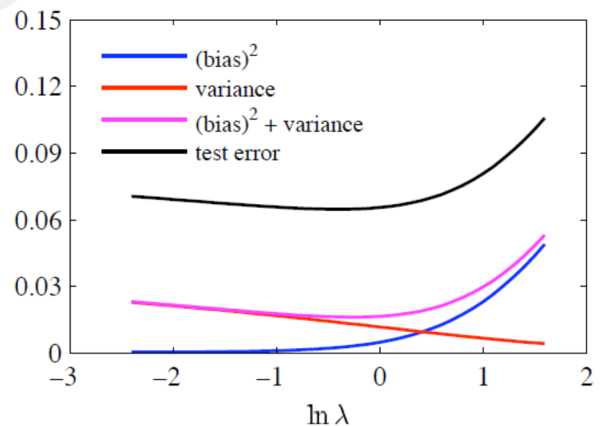
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Example

$$\text{average} = \bar{y}(x) = \frac{1}{L} \sum_{l=1}^L y^{(l)}(x)$$

$$(\text{bias})^2 = \frac{1}{N} \sum_{n=1}^N \{\bar{y}(x_n) - h(x_n)\}^2$$

$$\text{variance} = \frac{1}{N} \sum_{n=1}^N \frac{1}{L} \sum_{l=1}^L \{y^{(l)}(x_n) - \bar{y}(x_n)\}^2$$



Plot of squared bias and variance together with their sum. The minimum is at $\lambda = -0.31$ which is close to the value yielding minimum test error

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Bayesian Linear Regression

- Bias-variance decomposition requires splitting data. Inefficient.
- Avoids overfitting of maximum likelihood
- Leads to automatic way of determining model complexity
- **Look quickly at Bayesian approach.**
- **Posterior = Likelihood * Prior**
- **Log of Posterior = Log Likelihood + Log of Prior**

- Define prior over weights using zero-mean Gaussian prior:

$$p(\mathbf{w}|\alpha) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$$

- Log of posterior is sum of log likelihood and log of prior:

$$\ln p(\mathbf{w}|\mathbf{t}) = -\frac{\beta}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(x_n))^2 - \frac{\alpha}{2} \mathbf{w}^T \mathbf{w} + \text{constant}$$

with quadratic regularization term $\lambda = \frac{\alpha}{\beta}$ in least squares sense.

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Sequential Bayesian Learning

- Consider simple input variable x , single target t and a linear model of form

$$y(x, \mathbf{w}) = w_0 + w_1 x$$
- Just two weights, can plot prior and posteriors
- Generate synthetic data using $f(x_n, \mathbf{a}) = -0.3 + 0.5x_n + \epsilon$
- Goal is to recover $\mathbf{a} = \{-0.3, 0.5\}$ from data
- Basic algorithm:
 - Observe point (x, t) from dataset
 - calculate likelihood $p(t|x, \mathbf{w})$ based on estimate of noise precision β
 - multiply likelihood by previous prior over $\mathbf{w}$ to yield new posterior
 - Observe another point and continue

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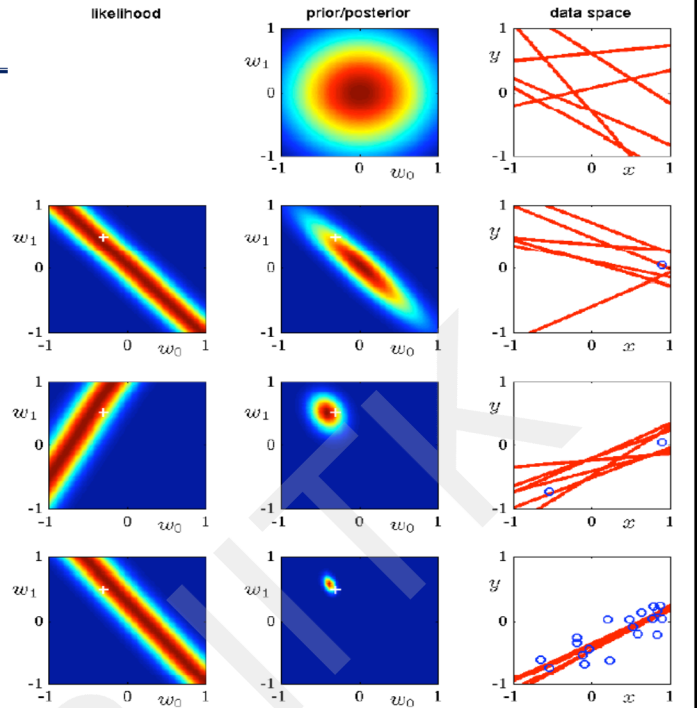
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Sequential Bayesian Learning

- Basic algorithm:
 - Observe point (x, t) from dataset
 - calculate likelihood $p(t|x, \mathbf{w})$ based on estimate of noise precision β
 - multiply likelihood by previous prior over $\mathbf{w}$ to yield new posterior
 - Observe another point and continue



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